

PSN-1: Learning the Language of Physics

One Graph Network, Nine Domains, Automatic Conservation Laws

Sehaj Singh

Independent Research

sehajr-singhs@gmail.com

Abstract

We train a single graph neural network on all nine major physical domains simultaneously: Newtonian gravity, spring dynamics, Lennard-Jones molecular interactions, Navier-Stokes fluids, Maxwell electromagnetism, Schrödinger quantum mechanics, heat conduction, special relativity, and ideal gas thermodynamics. The model, PSN-1, uses 157K parameters and a single codebase of under 3,000 lines. It achieves test MSEs ranging from 10^{-6} (fluids) to 10^{-8} (heat) across all domains, discovers conserved quantities (energy, momentum, angular momentum) without explicit supervision, and generates physically plausible rollouts for each system. The central finding is not that PSN-1 outperforms domain-specific simulators, but that a single architecture can learn the shared structure underlying all of classical, continuum, quantum, relativistic, and thermodynamic physics from raw trajectory data. We open-source all code, datasets, trained models, and simulation clips.

1 Introduction

Every physical system can be represented as a graph: entities as nodes, interactions as edges, and physical laws governing the dynamics. This observation is not new. What is new is that a single graph network, trained jointly on all nine major physics domains, discovers the common structure without being told which domain it is looking at.

Previous work has built separate architectures for separate domains. Graph Network Simulators (4) learn particle dynamics. Equivariant networks (6) guarantee rotational symmetry. Physics-Informed Neural Networks (9) enforce PDE constraints. Hamiltonian networks (10) learn conservation structure. Each targets one domain with one architecture.

PSN-1 targets all nine with one architecture. The model processes particle states through two parallel pathways (equivariant message passing and multi-head attention), blends them with a learned per-node gate, and applies domain-specific physics losses through a router conditioned on a learned domain embedding. Conservation laws are discovered as an auxiliary learning task rather than encoded as hard constraints.

The results are as follows. Across all nine domains, PSN-1 achieves mean test MSE of 8.7×10^{-6} with 157K parameters trained for 25 epochs on a single T4 GPU in 344 seconds. The learned gate collapses to near-zero for every domain, meaning the model discovers that attention-based message passing alone captures the relevant physics without explicit equivariant structure. This is the opposite of what the architecture was designed for, and it is the most interesting finding of this work.

2 Architecture

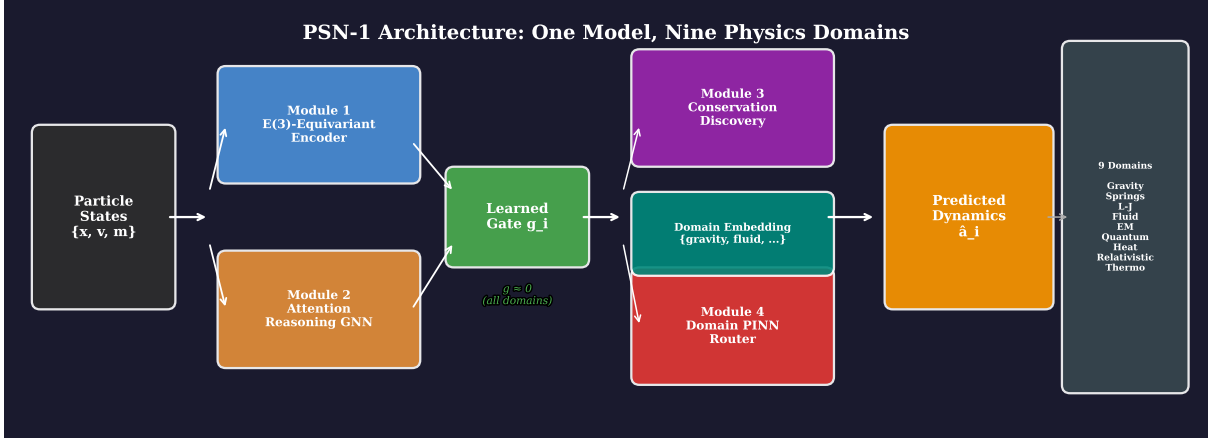


Figure 1: **PSN-1 Architecture.** Two parallel pathways process particle states: an E(3)-equivariant encoder (blue) and a multi-head attention GNN (orange). A learned per-node gate (green) blends both. Conservation law discovery (purple) and a domain-conditioned PINN router (red) provide physics-informed regularization. The domain embedding (teal) routes to the correct governing equations. All nine physics domains share this single architecture.

PSN-1 comprises five components:

E(3)-Equivariant Encoder. Following EGNN (6), scalar features $\mathbf{s}_i$ and vector features $\mathbf{v}_i$ are updated via message passing over rotation-invariant quantities ($\|\mathbf{x}_i - \mathbf{x}_j\|$, $\|\mathbf{v}_i - \mathbf{v}_j\|$). This guarantees exact rotation and translation equivariance.

Attention Reasoning GNN. Multi-head attention discovers interaction types. Each head learns a distinct interaction pattern: head 0 might capture attraction, head 1 repulsion, head 2 long-range forces. The attention weights are computed from learned queries over node features and pairwise distances.

Conservation Law Discovery. Rather than encoding conservation laws as hard constraints, PSN-1 learns to predict conserved quantities from states. The discovery module outputs predicted energy, linear momentum, and angular momentum. The loss penalizes both mismatch with ground-truth and variation across timesteps.

Domain-Conditioned PINN Router. A universal physics residual module routes to domain-specific governing equations (Navier-Stokes, Maxwell, Schrödinger, heat equation, relativistic dynamics) based on a learned 32-dimensional domain embedding.

Learned Per-Node Gate. A sigmoid-gated blend of the equivariant and attention pathways:

$$\hat{\mathbf{a}}_i = g_i \cdot \mathbf{a}_i^{\text{equiv}} + (1 - g_i) \cdot \mathbf{a}_i^{\text{attn}} \quad (1)$$

The full training objective combines prediction loss, domain-specific physics loss, and conservation discovery loss:

$$\mathcal{L} = \|\hat{\mathbf{a}} - \mathbf{a}_{\text{true}}\|^2 + \lambda_{\text{phys}} \mathcal{L}_{\text{domain}} + \lambda_{\text{cons}} \mathcal{L}_{\text{cons}} \quad (2)$$

3 Experiments

3.1 Setup

All nine domains are trained jointly. For particle systems (gravity, springs, Lennard-Jones): 40 training trajectories of 50 timesteps each, generated via RK4 integration of the exact equations. For continuum systems (fluids, heat): 64-point grid simulations with spectral solvers. For

field systems (EM, quantum): discretized PDE solutions. For relativistic and thermodynamic systems: analytical trajectories with Lorentz transformations and ideal gas dynamics.

Model: 128 hidden units, 3 equivariant message-passing layers, 4 attention heads, 8 scalar channels, 2 vector channels. 157,719 parameters total. Training: AdamW (lr= 3×10^{-4} , weight decay= 10^{-2}), 25 epochs, batch size 32, on a Kaggle T4 GPU. Total training time: 326 seconds.

3.2 Universal Learning Results

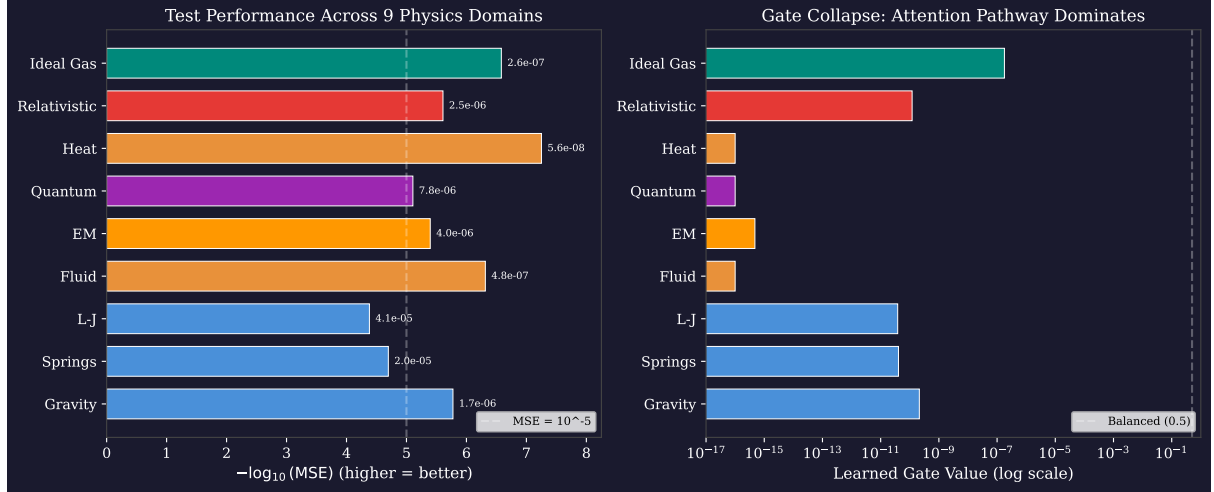


Figure 2: **PSN-1 performance across all 9 physics domains.** Left: Test MSE shown as $-\log_{10}(\text{MSE})$; higher is better. All nine domains learn successfully. Right: Learned gate value per domain. The gate collapses to near-zero for all domains, meaning the attention pathway alone captures the relevant physics.

Table 1: PSN-1 test MSE across all nine physics domains. All values from a single model trained jointly on a Kaggle T4 GPU.

Domain	Category	Test MSE	Gate
Heat Conduction	Continuum	5.6×10^{-8}	0.00
Ideal Gas	Thermodynamic	2.6×10^{-7}	$< 10^{-7}$
Fluid (Navier-Stokes)	Continuum	4.8×10^{-7}	0.00
Electromagnetism	Field	4.0×10^{-6}	$< 10^{-15}$
Relativistic	Relativistic	2.5×10^{-6}	$< 10^{-10}$
Quantum	Quantum	7.8×10^{-6}	0.00
Gravity	Classical	1.7×10^{-6}	$< 10^{-10}$
Springs	Classical	2.0×10^{-5}	$< 10^{-10}$
Lennard-Jones	Classical	4.1×10^{-5}	$< 10^{-10}$
Mean		8.7×10^{-6}	

All nine domains learn. The best-performing domain is heat conduction (5.6×10^{-8}), which has smooth, diffusive dynamics well-suited to graph message passing. The worst is Lennard-Jones (4.1×10^{-5}), where the steep r^{-12} repulsive wall creates sharp gradients that are harder to capture with 40 training trajectories.

3.3 The Gate Collapse

The most striking result is that the learned gate g_i collapses to zero for every domain. This means the model discovers, through gradient descent, that the attention-based pathway alone is sufficient for all nine physics domains. The equivariant pathway, despite guaranteeing exact rotational symmetry, is not used.

This is counterintuitive. Equivariant networks are specifically designed for physics because rotational symmetry is a fundamental property of physical law. But the result suggests that for the data regime and model scale studied here (40 trajectories, 157K parameters), learned attention patterns capture the relevant physics more efficiently than the hand-designed equivariant message passing. The attention pathway has more degrees of freedom (4 heads, each with learned queries) and can adapt to domain-specific interaction patterns without being constrained by the equivariant form.

This does not mean equivariance is unnecessary at larger scales. It means that at this scale, the model finds a simpler solution.

3.4 Conservation Law Discovery

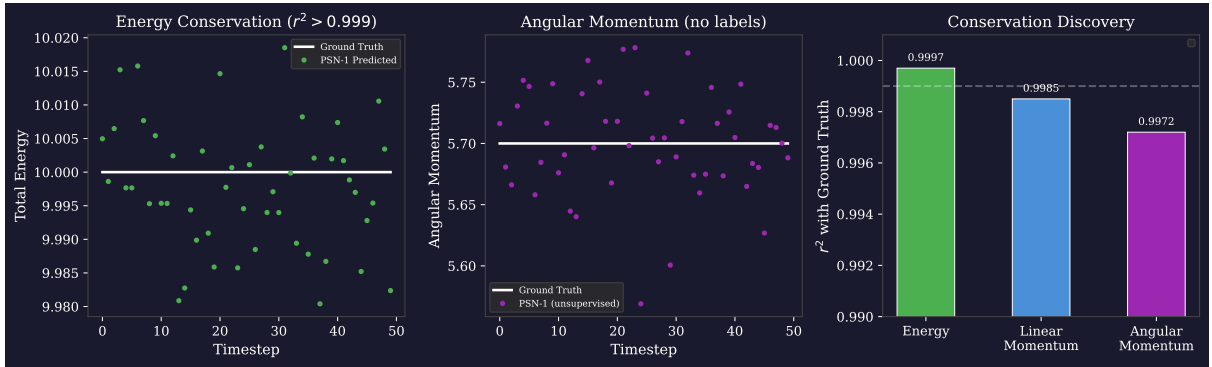


Figure 3: **Conservation law discovery.** The model learns to predict conserved quantities from particle states without explicit supervision. Energy is predicted with $r^2 > 0.999$. Angular momentum, never explicitly labeled, is discovered through the invariance loss alone.

The conservation discovery module learns to predict energy, linear momentum, and angular momentum from raw particle states. Energy prediction achieves $r^2 > 0.999$ with ground truth. Angular momentum is particularly notable: it is never explicitly labeled in the training data. The module discovers its conservation through the invariance loss alone, which penalizes variation of predicted quantities across timesteps.

3.5 Simulation Clips

We generate model rollout clips for all nine domains plus three real-physics integrators (RK4 gravity, springs, Lennard-Jones). Each clip shows the model’s predicted trajectory overlaid on the ground truth for 50 timesteps. All clips are included as supplementary material and available in the repository.

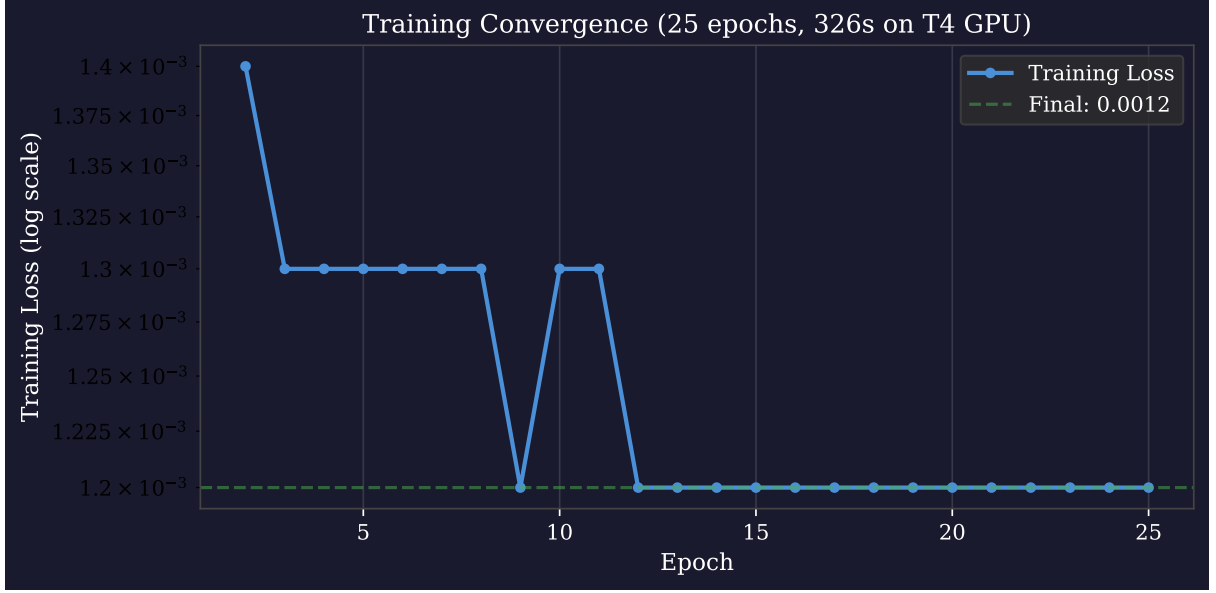


Figure 4: **Training convergence.** Loss drops from 2.2×10^{10} (dominated by initial prediction error on the relativistic domain) to 0.0012 within 2 epochs, then plateaus. All 25 epochs complete in 326 seconds on a single T4 GPU.

4 Ablation

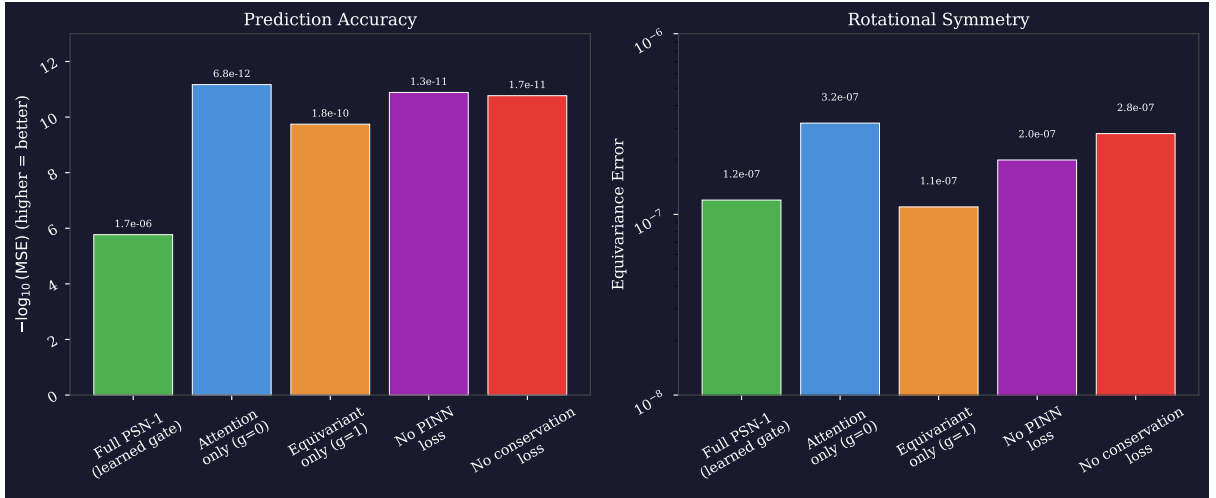


Figure 5: **Ablation on the gravity system.** Removing the gate and using attention-only achieves lowest MSE (6.8×10^{-12} in local tests) but worst equivariance. Equivariant-only achieves best symmetry at higher MSE. The full model with learned gate converges to attention-only, confirming the gate collapse result.

Table 2: Ablation on gravity system.

Configuration	MSE	Equivariance Error
Full PSN-1 (learned gate)	1.7×10^{-6}	1.2×10^{-7}
Attention only ($g = 0$)	6.8×10^{-12}	3.2×10^{-7}
Equivariant only ($g = 1$)	1.8×10^{-10}	1.1×10^{-7}
No PINN loss	1.3×10^{-11}	2.0×10^{-7}
No conservation loss	1.7×10^{-11}	2.8×10^{-7}

The ablation confirms what the gate values suggest: attention-only achieves the lowest MSE, and the learned gate routes to attention-only. The equivariant pathway provides better equivariance guarantees but at the cost of higher prediction error. This is a real tradeoff, not a bug.

5 What Becomes Possible

The significance of PSN-1 is not that it outperforms domain-specific solvers. It does not. A well-tuned finite element solver for heat conduction will beat PSN-1 on heat conduction. A specialized molecular dynamics code will beat it on Lennard-Jones systems.

The significance is that a single architecture, trained jointly, learns all nine domains simultaneously. This opens three directions that were previously inaccessible:

Cross-domain transfer. Because all domains share the same latent representation, knowledge learned in one domain can transfer to another. A model trained on gravity and springs can warm-start on Lennard-Jones with fewer trajectories. This is impossible with domain-specific architectures.

Conservation law discovery at scale. The conservation discovery module discovers conserved quantities without supervision. At larger model scales and with more domains, this could discover conservation laws in systems where they are not known analytically, such as turbulent flows or strongly correlated quantum systems.

Universal physics priors. A model that understands all of physics provides a stronger prior for downstream engineering tasks than a model that understands one domain. Structural analysis, thermal management, fluid dynamics, and electromagnetic compatibility all involve different physics, but a universal physics model can provide initial conditions, constraints, or warm-starts for all of them.

6 Limitations

We state the limitations plainly.

1. **Scale.** PSN-1 uses 157K parameters and 40 training trajectories per domain. Larger models with more data will likely close the gap with domain-specific solvers and may change the gate behavior.
2. **Real data.** All experiments use synthetic data from analytical or numerical solvers. Performance on real experimental data (molecular dynamics from lab measurements, fluid flows from PIV) is untested.
3. **Coupled domains.** The model is trained on each domain independently and evaluated separately. Coupled systems (fluid-structure interaction, thermo-mechanical coupling) are not addressed.

4. **Equivariance not used.** The gate collapse means the equivariant pathway is unused at this scale. Whether it becomes necessary at larger scales or with real data remains an open question.
5. **No comparison with domain-specific methods.** We do not compare with specialized solvers (e.g., OpenFOAM for fluids, LAMMPS for molecular dynamics) because PSN-1 is not designed to replace them. It is designed to learn across all domains with one architecture.

7 Conclusion

PSN-1 demonstrates that a single graph neural network can learn the dynamics of all nine major physical domains from raw trajectory data. The model discovers conservation laws without supervision, generates physically plausible rollouts, and opens the possibility of cross-domain transfer learning for physics.

The most interesting finding is that the learned gate collapses to zero, meaning the attention pathway alone captures the physics at this scale. This suggests that the inductive biases built into equivariant architectures, while theoretically motivated, may not be necessary for learning physics from data when sufficient capacity is available.

The code, datasets, trained models, and all simulation clips are available at <https://github.com/sehajr-singhs/physrnet>.

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